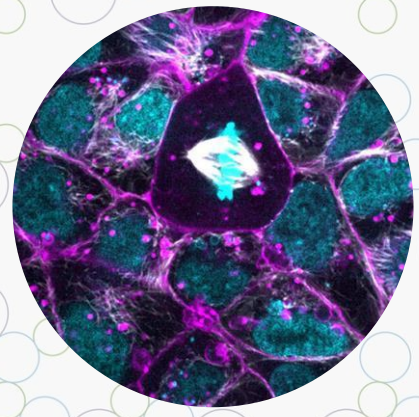


Basic Python Part 2 - Control Flow and Functions

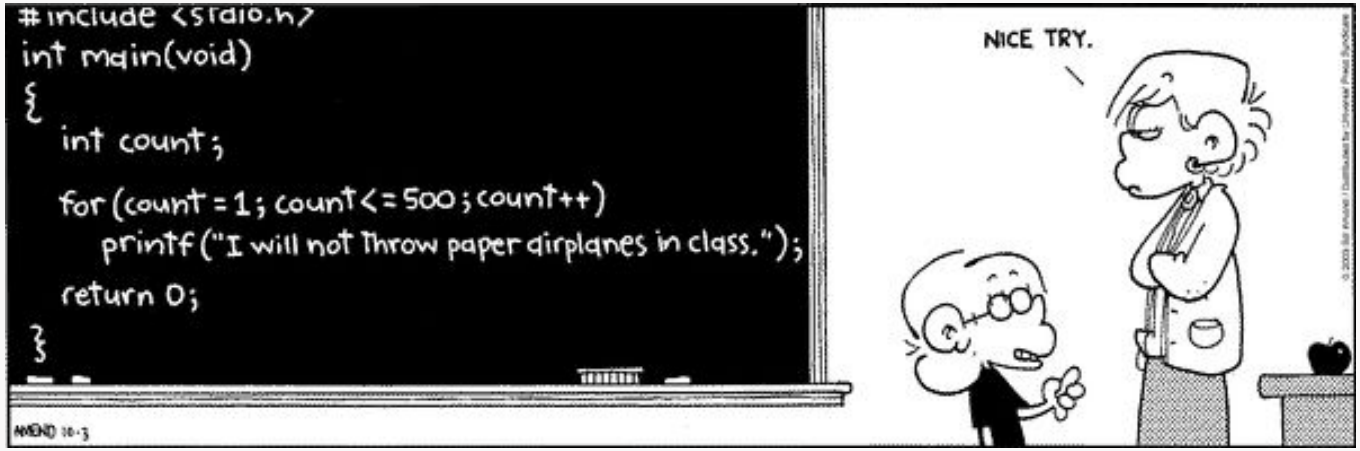
Coding with Allen Institute Data: A Python Workshop for Undergraduate Neuroscience Educators

Objectives for this session

- Write a **for loop** to iterate over elements in an object
- Recognize **Booleans** & write conditional logic statements
- Test **conditional statements** in Python
- Recognize **function** syntax & write a simple function

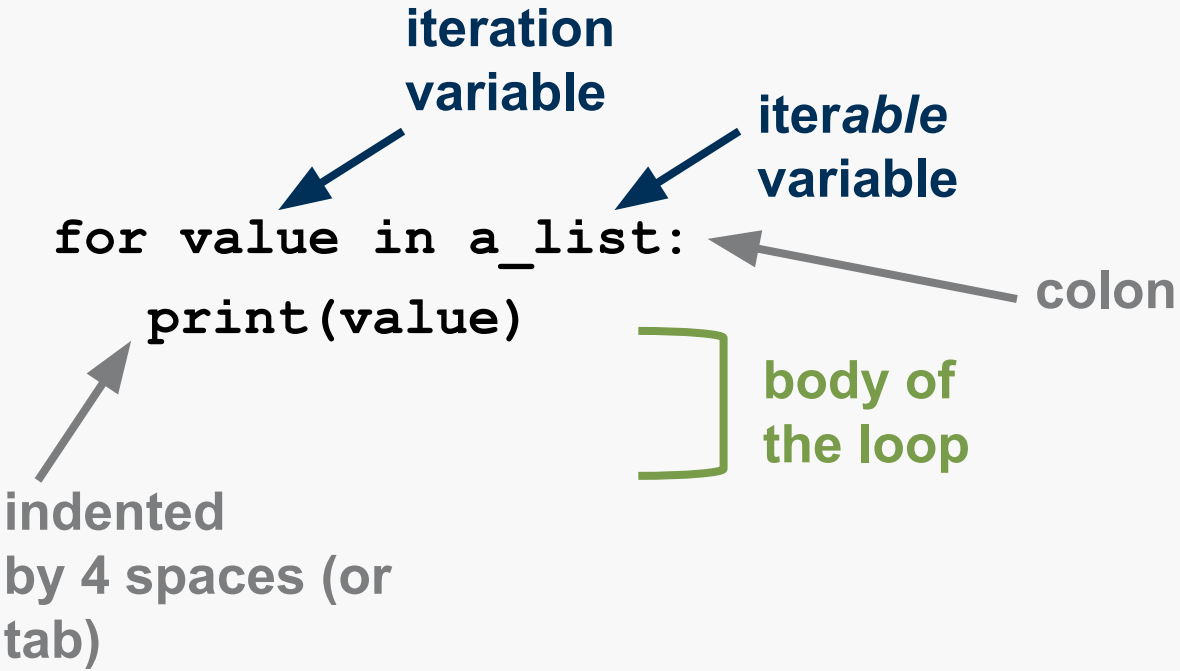
A loop is a procedure to repeat a piece of code. (another way of saying this is that it iterates through code)

- Loops enable you to re-run blocks of code as many times as you need.
- Python has two main ways to run loops: `for` & `while`



Bill Amend, FoxTrot, October 3, 2003

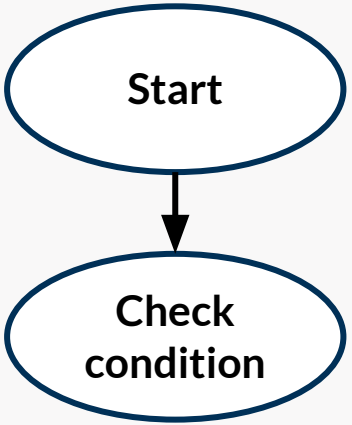
`for` loop syntax



A **for loop** is a procedure to repeat code for every element in a sequence.

`for` loop syntax

```
a_list = [1,2,3]
for value in a_list:
    print(value)
```



`for` loop syntax

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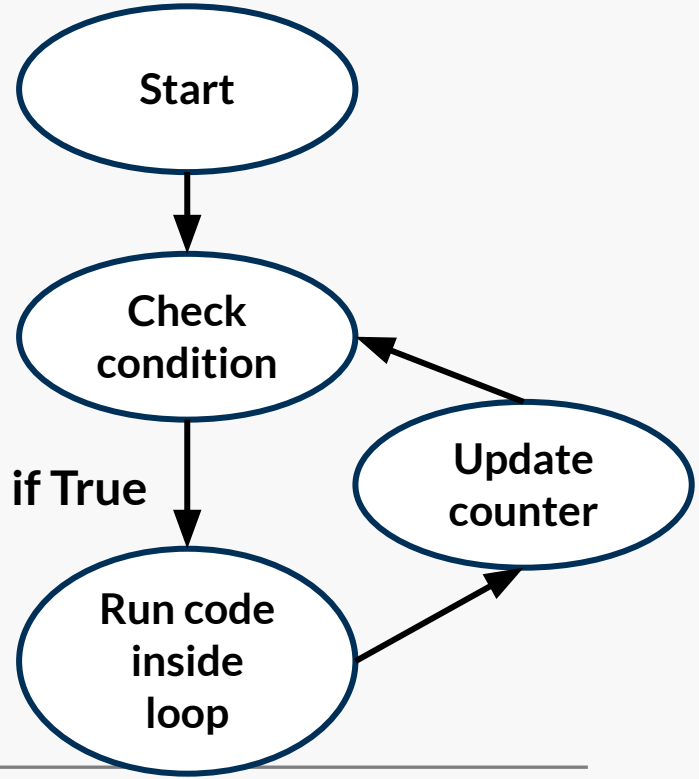
a_list = [1,2,3]
for value in a_list:
    print(value)
```

```
1
2
3
```

output



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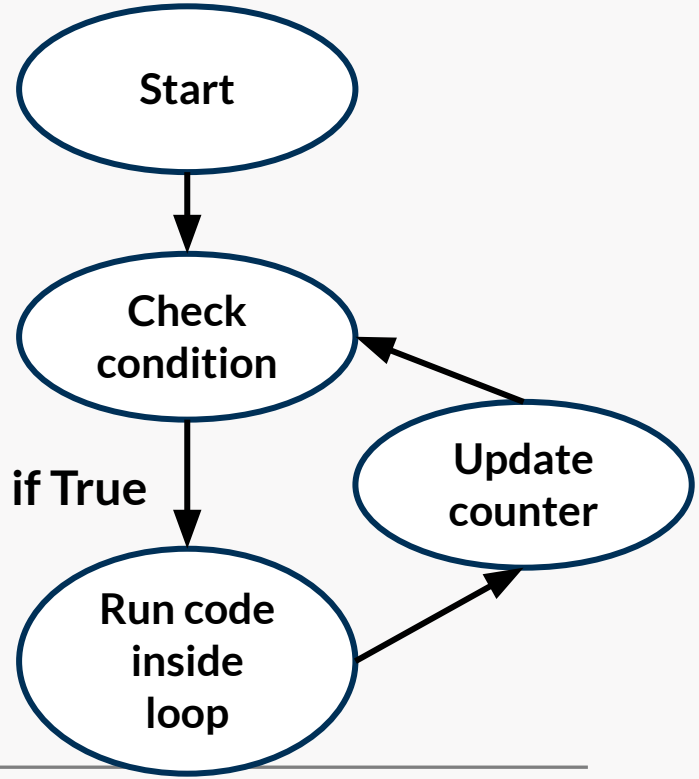

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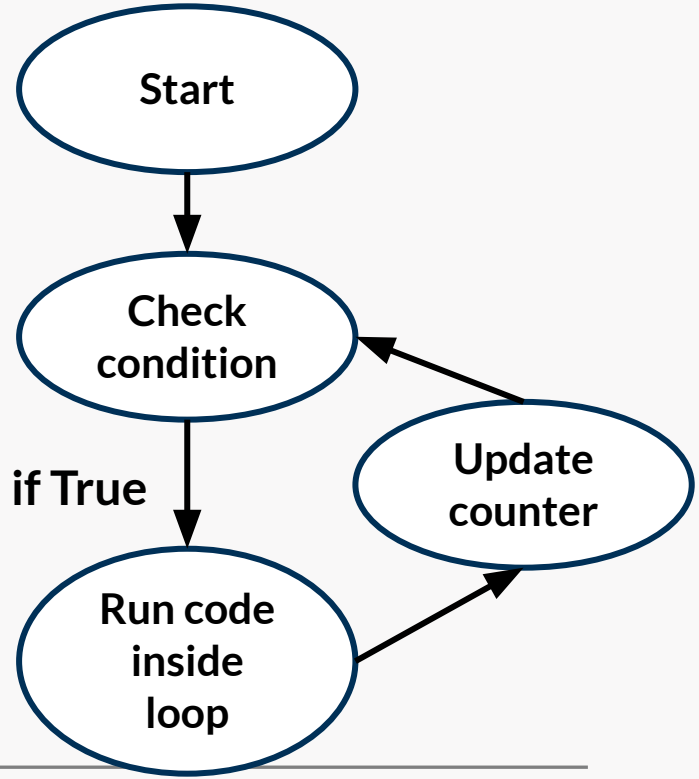

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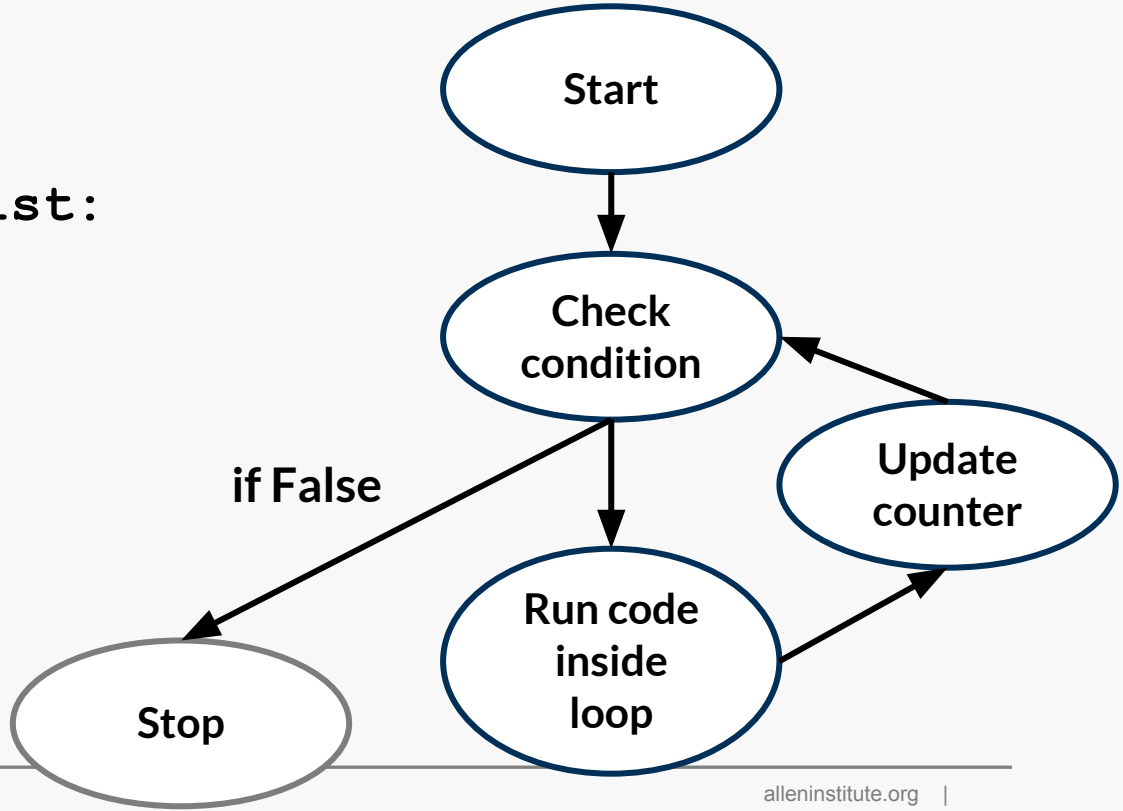


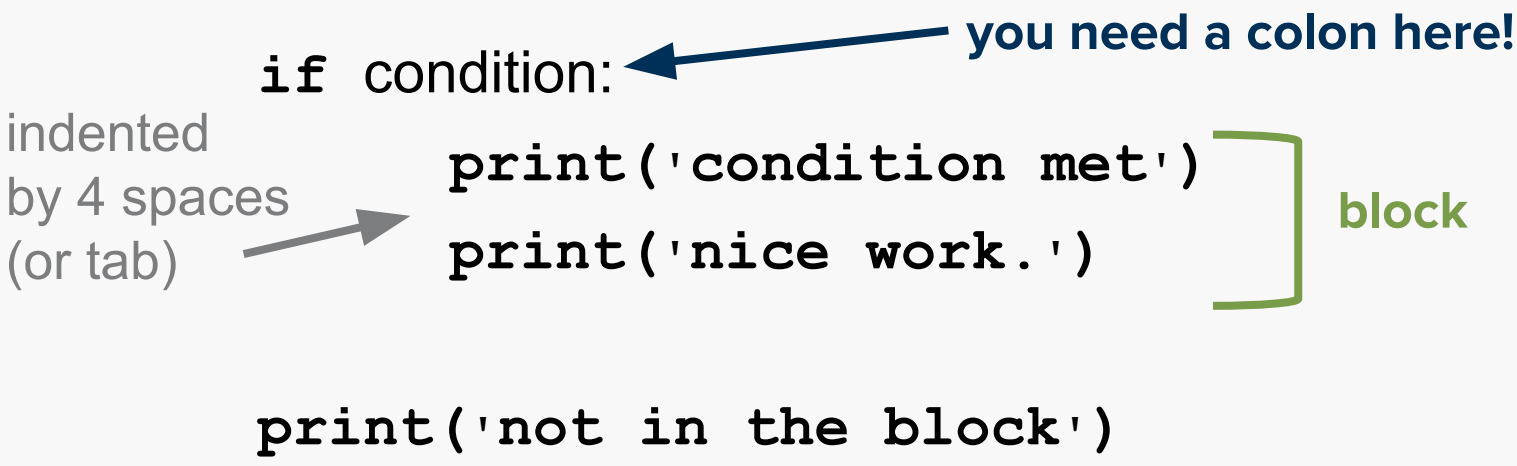
Basic conditional operators in Python

Symbol	Operation	Usage	Outcome
==	Is equal to	`10==5*2`	`True`
!=	Is not equal to	`10 != 5*2`	`False`
>	Is greater than	`10 > 2`	`True`
<	Is less than	`10 < 2`	`False`
>=	Greater than *or* equal to	`10 >= 10`	`True`
<=	Less than *or* equal to	`10 <= 10`	`True`

Boolean variables store `True` (1) or `False` (0) and are the basis of all computer operations.

Syndey Padua:
https://sydneypadua.com/2dgoggles/happy-200th-birthday-george-boole/

`if` statements syntax



`if/else` statement syntax

```
if condition:
        print('condition met')
        print('nice work.')
else:
        print('condition not met')
```

you need a colon here!

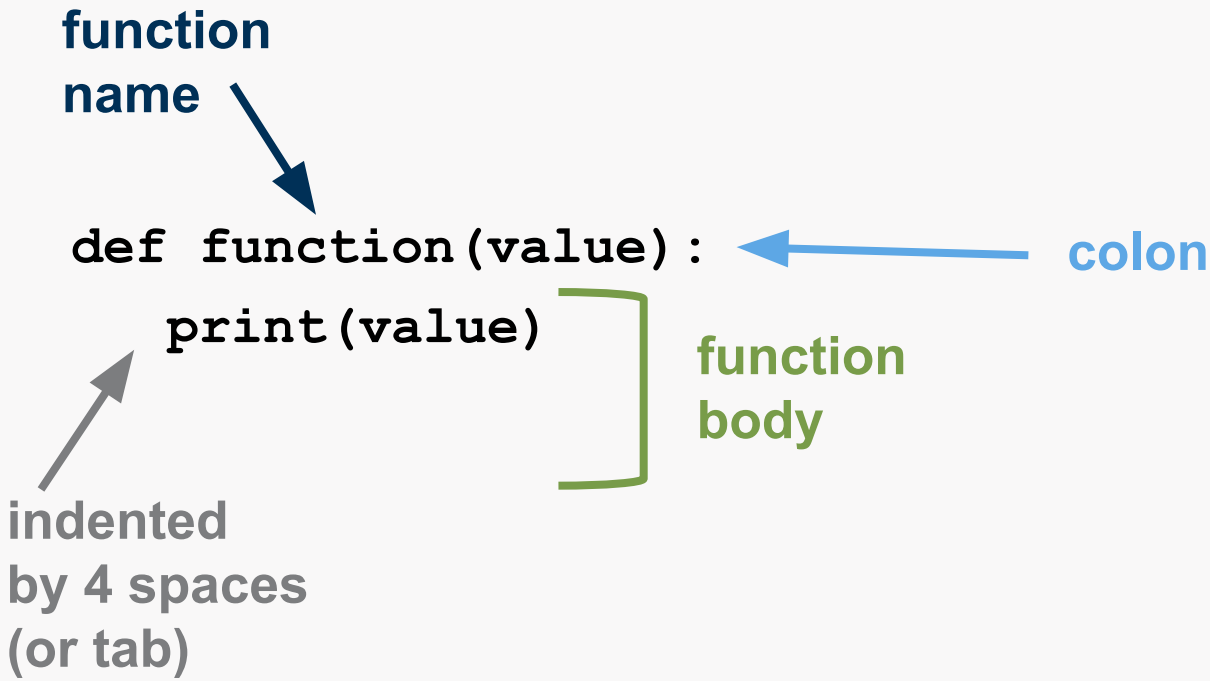
Functions

Functions are pieces of code that are designed to do one task

Functions take in inputs, process those inputs, and possibly return an output.

Python has built-in functions, but we can also write our own!

`function` syntax



`function` syntax

input arguments (these can be variables or default arguments)

```
def function(b):
    a = b**2
    return a
```

return to retrieve a variable outside of a function *(what happens in the function stays in the function)* ALSO ENDS THE FUNCTION!

call to function giving it the argument and saving the returned variable as a

```
a = function(6)
```

Resources

Intro. to Comp. Sci. & OOP: Python · Cogniterra

Plotting and Programming in Python: For Loops

Plotting and Programming in Python: Conditionals

Whirlwind Tour of Python: Control Flow

Merely Useful Functions

Python Tutorial: Functions